

as Specht technique in transformer engineering books, is used to calculate the peak value of inrush current for selected number of cycles. The procedure is as follows:

Step 1. Inrush current is calculated for the worst-case condition, considering that the transformer is energised at the instant when voltage is zero and residual flux is in the same direction as the initial flux change. This gives the maximum possible value of inrush current.

Once the core saturates, inrush current gets limited by air-core reactance X_s , which is calculated as:

$$X_s = \frac{\mu_0 \times N^2 \times A_w}{h_w} \times 2.0 \times \pi \times f$$

where N is the number of turns of the excited winding, μ_0 permeability of free space, A_w area inside the mean turn of excited winding $\{ = (\pi/4) \times (\text{mean diameter})^2 \}$ and h_w height of the energised winding.

Step 2. Angle θ at which the core saturates is given as:

$$\theta = k1 \times \cos^{-1} \left(\frac{Bs - B_m}{Br - B_m} \right)$$

where B_m is peak flux density in Tesla, Br residual flux density in Tesla and Bs saturation flux density of the core material in Tesla.

For a core made of cold rolled steel, $Br = 80\%$ of B_m

For hot rolled steel, $Br = 60\%$ of B_m
 $k1 =$ correction factor for saturation angle

That is, $k1 = 0.9$

Step 3. Inrush current peak for the first cycle on single-phase basis is calculated as:

$$i_{\text{omas}} = \frac{k2 \times V/\sqrt{2}}{X_s} \times (1 - \cos \theta)$$

where V is rms phase value of the primary/excited winding and $k2$ correction factor for the peak value ($= 1.15$).

Step 4. Residual flux density at the end of the first cycle is calculated in this step. The residual component of flux density reduces due to damping in the circuit provided by the resistance of the system. The new value of residual flux density is calculated as:

$$Br(\text{new}) = Br(\text{old}) - \frac{(B_m \times k3 \times R) \times [2 \times \sin(\theta) - \theta \times \cos(\theta)]}{X_s}$$

where $k3 =$ correction factor for the decay of inrush $= 2.26$

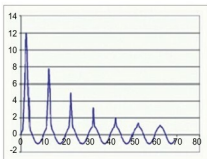


Fig. 3: Typical inrush current waveform

Steps 2 through 4 are repeated to calculate the peak value of inrush current for all the subsequent cycles.

Effect of switching angle on the peak value of inrush current

The key factor in determining the magnitude of the magnetising inrush current is the switching angle of voltage source. The following formula, which takes into account the effect of switching angle and circuit resistance to predict the first peak of inrush current, has been used:

$$i_{\text{inrush}} = \frac{(V_{\text{peak}})}{\sqrt{R^2 + X_s^2}} \times \left\{ \frac{(Br - Bs)}{B_{\text{peak}}} + \cos(\theta) + 1 \right\}$$

where R is the total resistance including system resistance and θ is the initial phase angle of the voltage source.

Features of the software

The software presented here performs inrush current analysis of either a single-phase or three-phase transformer using the above-mentioned mathematical model and predicts the following characteristics:

1. Calculates the peak value of inrush current for individual cycles and plots them on a graph, thus providing inrush current envelope
2. Calculates the effect of switching angle on the peak value of inrush current
3. Calculates the magnitude of harmonics of different orders generated due to inrush current and determines total harmonic distortion (THD) due to inrush current

The software presented here is developed in Java Standard Edition (SE) using the software development kit (SDK) version Java 2 SDK, SE v1.8.0_60.

The software is user-friendly

with the front end as a GUI, which is a Java Swing based class through which the user interacts and selects various options to run the program and enter the data. At the back end, these options are processed in the code written in core Java. Results are displayed using Java Swing features. That is, inrush current graph is plotted using graphics feature of Swing and results of the effect of switching angle and harmonic analysis are displayed in tables.

The program uses the following files:

inrush_4a.java. This file contains following classes:

inrush_4a. This is the main class. It is used to accept options from the user and process them. Options include Enter Transformer Data, Display Inrush Current Graph, Calculate Effect of Switching Angle and Harmonic Analysis.

EvenOddRenderer.class. This class is used to set properties of the table in which the result of effect of switching angle and harmonic analysis will be displayed.

Main.class. In this software, a provision has been made to allot an ID number to each transformer job. The generateRandomString() function of main.class is used to generate the transformer ID number at the time of entering transformer data.

LineChart_use.java. This file contains the class LineChart_use, which facilitates plotting of inrush current graph.

Steps to run the program

In order to run the program, batch file inrush4a.bat is created. Batch file is an executable file. It contains a set of commands to run the program. The advantage of using the batch file is that you need not set the address of the directory, files and other resource files again each time you run the program. These issues are handled by batch file.

inrush4a.bat batch file contains the following commands:

```
@ECHO OFF
set path=C:\Program Files (x86)\Java\
jdk1.8.0_60\bin;%PATH%;%path%
javac inrush_4a.java LineChart_use.java
java inrush_4a
```